Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Given :
(I) $\mathrm{H_2(\ g) + \frac{1}{2}O_2(\ g) \rightarrow H_2O(}l\mathrm{)}$
$\Delta \mathrm{H^\circ_{298\ K}} = -285.9\ \mathrm{kJ\ mol^{-1}}$
(II) $\mathrm{H_2(\ g) + \frac{1}{2}O_2(\ g) \rightarrow H_2O(g)}$
$\Delta \mathrm{H^\circ_{298\ K}} = -241.8\ \mathrm{kJ\ mol^{-1}}$
The molar enthalpy of vapourisation of water will be :

(A) $241.8\ \mathrm{kJ\ mol^{-1}}$

(B) $22.0\ \mathrm{kJ\ mol^{-1}}$

(C) $44.1\ \mathrm{kJ\ mol^{-1}}$

(D) $527.7\ \mathrm{kJ\ mol^{-1}}$

Q.2

Given Reaction Energy Change (in $\mathbf{kJ}$) $\mathrm{Li(s) \rightarrow Li(g)}$
161
$\mathrm{Li(g) \rightarrow Li^+(g)}$
520
$\mathrm{\frac{1}{2}\ F_2(\ g) \rightarrow F(g)} \quad 77$
$\mathrm{F(g) + e^- \rightarrow F^-(g)}$
(Electron gain enthalpy) $\mathrm{Li^+(g) + F^-(g) \rightarrow LiF(s) - 1047}$
$\mathrm{Li(s) + \frac{1}{2}\ F_2(\ g) \rightarrow LiF(s) - 617}$
Based on data provided, the value of electron gain enthalpy of fluorine would be :

(A) $-300\ \mathrm{kJ\ mol^{-1}}$

(B) $-350\ \mathrm{kJ\ mol^{-1}}$

(C) $-328\ \mathrm{kJ\ mol^{-1}}$

(D) $-228\ \mathrm{kJ\ mol^{-1}}$

Q.3

Hydrogen has an ionisation energy of $1311\ \mathrm{kJ\ mol^{-1}}$ and for chlorine it is $1256\ \mathrm{kJ\ mol^{-1}}$. Hydrogen forms $\mathrm{H^+}$ (aq) ions but chlorine does not form $\mathrm{Cl^+(aq)}$ ions because

(A) $\mathrm{H^+}$ has lower hydration enthalpy

(B) $\mathrm{Cl^+}$ has lower hydration enthalpy

(C) Cl has high electron affinity

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(D) Cl has high electronegativity

Q.4

At $25^{\circ} \mathrm{C}$, when 1 mole of MgSO_4 was dissolved in water, the heat evolved was found to be $91.2 \mathrm{~kJ}$. One mole of $\mathrm{MgSO}_4.7\mathrm{H}_2\mathrm{O}$ on
dissolution gives a solution of the same composition accompanied by an absorption of $13.8 \mathrm{~kJ}$. The enthalpy of hydration, i.e., $\Delta \mathrm{H}_{\mathrm{h}}$ for the reaction
$\mathrm{MgSO}_4(\mathrm{~s})+7 \mathrm{H}_2 \mathrm{O}(\mathrm{l}) \longrightarrow \mathrm{MgSO}_4.7 \mathrm{H}_2 \mathrm{O}(\mathrm{s})$
is:

(A) $-105 \mathrm{~kJ} / \mathrm{mol}$

(B) $-77.4 \mathrm{~kJ} / \mathrm{mol}$

(C) $105 \mathrm{~kJ} / \mathrm{mol}$

(D) None of these

Q.5

Calculate the heat produced (in kJ) when $224 \mathrm{~g}$ of CaO is completely converted to CaCO_3 by reaction with CO_2 at $27^{\circ} \mathrm{C}$ in a container of fixed volume.
Given : $\Delta \mathrm{H}_f^{\circ}\left(\mathrm{CaCO}_3,\ \mathrm{s}\right)=-1207 \mathrm{~kJ} / \mathrm{mol}$
$\Delta \mathrm{H}_f^{\circ}(\mathrm{CaO}, \mathrm{s})=-635 \mathrm{~kJ} / \mathrm{mol}, \Delta \mathrm{H}_f^{\circ}\left(\mathrm{CO}_2,\ \mathrm{g}\right)=$
$-394 \mathrm{~kJ} / \mathrm{mol}; \quad\left[\text { Use } \mathrm{R}=8.3 \mathrm{JK}^{-1} \mathrm{~mol}^{-1}\right]$

(A) $702.04 \mathrm{~kJ}$

(B) $721.96 \mathrm{~kJ}$

(C) $712 \mathrm{~kJ}$

(D) $721 \mathrm{~kJ}$

Q.6

The amount of energy released when $20 \mathrm{~mL}$ of $0.5 \mathrm{MNaOH}$ are mixed with $100 \mathrm{~mL}$ of $0.1 \mathrm{M}$ HCl is xkJ. The heat of neutralisation (in $\mathrm{kJ~mol}^{-1}$) is

(A) $-100 \mathrm{x}$

(B) $-50 \mathrm{x}$

(C) $+100 \mathrm{x}$

(D) $+50 x$

Q.7

Which of the following value of $\Delta_f \mathrm{H}^{\circ}$ represent that the product is least stable ?

(A) $-94.0 \mathrm{kcalmol}^{-1}$

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(B) $-231.6 \mathrm{kcal} \mathrm{mol}^{-1}$

(C) $+21.4 \mathrm{kcal} \mathrm{mol}^{-1}$

(D) $+64.8 \mathrm{kcal} \mathrm{mol}^{-1}$

Q.8

Enthalpy of neutralisation of HCl with NaOH is x. The heat evolved when $500 \mathrm{~mL}$ of $2 \mathrm{~N} \mathrm{HCl}$ are mixed with $250 \mathrm{~mL}$ of $4 \mathrm{~N} \mathrm{NaOH}$ will be.

(A) $500 \mathrm{x}$

(B) $100 x$

(C) x

(D) $10 \mathrm{x}$

Q.9

The enthalpy of neutralisation of $\mathrm{NH}_4\mathrm{OH}$ with HCl is $-51.46 \mathrm{~kJ} \mathrm{~mol}^{-1}$ and the enthalpy of neutralisation of NaOH with HCl is $-55.90 \mathrm{~kJ} \mathrm{mol}^{-1}$. The enthalpy of ionisation of $\mathrm{NH}_4\mathrm{OH}$
is

(A) $-107.36 \mathrm{~kJ} \mathrm{~mol}^{-1}$

(B) $-4.44 \mathrm{~kJ} \mathrm{~mol}^{-1}$

(C) $+107.36 \mathrm{~kJ} \mathrm{~mol}^{-1}$

(D) $+4.44 \mathrm{~kJ} \mathrm{~mol}^{-1}$

Q.10

The enthalpies of formation of $\mathrm{Al}_2\mathrm{O}_3$ and $\mathrm{Cr}_2\mathrm{O}_3$ are $-1596 \mathrm{~kJ}$ and $-1134 \mathrm{~kJ}$ respectively. $\Delta \mathrm{H}$ for the reaction $2 \mathrm{Al}+\mathrm{Cr}_2\mathrm{O}_3 \rightarrow 2 \mathrm{Cr}+\mathrm{Al}_2\mathrm{O}_3$ is

(A) $-2730 \mathrm{~kJ}$

(B) $-462 \mathrm{~kJ}$

(C) $-1365 \mathrm{~kJ}$

(D) $+2730 \mathrm{~kJ}$

Q.11

Which of the following salts should cause maximum cooling when 1 mole of it is dissolved in the same volume of water ?

(A) $\mathrm{NaCl} ; \Delta \mathrm{H}^{\mathrm{o}}=5.35 \mathrm{~kJ} \mathrm{~mol}^{-1}$

(B) $\mathrm{KNO}_3 ; \Delta \mathrm{H}^{\circ}=53.5 \mathrm{~kJ} \mathrm{~mol}^{-1}$

(C) $\mathrm{KOH} ; \Delta \mathrm{H}^{\circ}=-56.0 \mathrm{~kJ} \mathrm{~mol}^{-1}$

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(D) $\mathrm{HBr}; \Delta \mathrm{H}^{\mathrm{o}}=-83.3 \mathrm{~kJ} \mathrm{~mol}^{-1}$

Q.12

The molar enthalpies of combustion of isobutane and n -butane are $-2870 \mathrm{~kJ} \mathrm{~mol}^{-1}$ and $-2878 \mathrm{~kJ} \mathrm{~mol}^{-1}$ respectively at $298 \mathrm{~K}$ and $1 \mathrm{~atm}$. Calculate $\Delta \mathrm{H}^{\circ}$ for the conversion of 1 mole of n-butane to 1 mole of isobutane

(A) $-8 \mathrm{~kJ} \mathrm{~mol}^{-1}$

(B) $+8 \mathrm{~kJ} \mathrm{~mol}^{-1}$

(C) $-5748 \mathrm{~kJ} \mathrm{~mol}^{-1}$

(D) $+5748 \mathrm{~kJ} \mathrm{~mol}^{-1}$

Q.13

The enthalpy of atomisation of CH_4 and $\mathrm{C}_2 \mathrm{H}_6$ are 360 and $620 \mathrm{kcalmol}^{-1}$ respectively. The $\mathrm{C}-\mathrm{C}$ bond energy is expected to be

(A) $210 \mathrm{kcalmol}^{-1}$

(B) $80 \mathrm{kcalmol}^{-1}$

(C) $130 \mathrm{kcalmol}^{-1}$

(D) $180 \mathrm{kcalmol}^{-1}$

Q.14

Boron can undergo the following reactions with the given enthalpy changes:
$2 \mathrm{~B}(\mathrm{~s})+\frac{3}{2} \mathrm{O}_2(\mathrm{~g}) \longrightarrow \mathrm{B}_2 \mathrm{O}_3(\mathrm{~s}); \Delta \mathrm{H}=-1260 \mathrm{~kJ}$
$2 \mathrm{~B}(\mathrm{~s})+3 \mathrm{H}_2(\mathrm{~g}) \longrightarrow \mathrm{B}_2 \mathrm{H}_6(\mathrm{~g}); \Delta \mathrm{H}=30 \mathrm{~kJ}$
Assume no other reactions are occurring. If in a container (operating at constant pressure) which is isolated from the surrounding, mixture of H_2 (gas) and O_2 (gas) are passed over excess of $\mathrm{B}(\mathrm{s})$, then calculate the molar ratio $\left(\mathrm{O}_2: \mathrm{H}_2\right)$ so that temperature of the container do no change :

(A) $15: 3$

(B) $1: 42$

(C) $42: 1$

(D) $1: 84$

Q.15

The temperature of 4 moles of an ideal gas is raised from $300 \mathrm{~K}$ to $350 \mathrm{~K}$. What is the vahe of $\Delta H-\Delta E$ for this process ? $\left(R=8.3 \mathrm{~J} \mathrm{~mol}^{-1} K^{-1}\right)$

(A) 0

(B) $415 \mathrm{~J}$

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(C) $41.5\ \mathrm{J}$

(D) $1660\ \mathrm{J}$

Q.16

The enthalpy change (ΔH) for the process $\mathrm{N_2H_4(\ g) \rightarrow 2\ N(\ g) + 4H(g)}$ is $1724\mathrm{KJmol^{-1}}$. If the bond energy of $\mathbf{N-H}$ bond in ammonia is $\mathbf{391KJmol^{-1}}$. What is the bond energy of $\mathrm{N-N}$ bond is $\mathbf{N_2H_4}$

(A) $160\mathrm{KJmol^{-1}}$

(B) $391\mathrm{KJmol^{-1}}$

(C) $1173\mathrm{KJmol^{-1}}$

(D) $320\mathrm{KJmol^{-1}}$

Q.17

Consider a spherical shell of radius R at temperature T. The black body radiation inside it can be considered as an ideal gas of photons with internal energy per unit volume $u = \frac{U}{V} \propto T^4$ and pressure $\mathrm{p} = \frac{1}{3}\left(\frac{\mathrm{U}}{\mathrm{V}}\right)$. If the shell now undergoes an adiabatic expansion the relation between T and R is :

(A) $\mathrm{T} \propto \frac{1}{\mathrm{R}}$

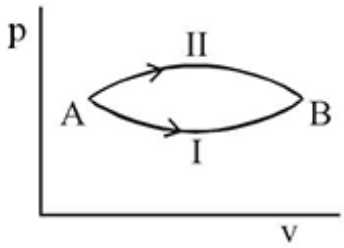
(B) $\mathrm{T} \propto \frac{1}{\mathrm{R}^3}$

(C) $\mathrm{T} \propto \mathrm{e}^{-\mathrm{R}}$

(D) $\mathrm{T} \propto \mathrm{e}^{-3\mathrm{R}}$

Q.18

A system goes from A to B via two processes I and II as shown in figure. If ΔU_1 and ΔU_2 are the changes in internal energies in the processes I and II respectively, then the relation between ΔU_1 and ΔU_2 is



(A) $\Delta U_1 = \Delta U_2$

(B) $\Delta U_2 < \Delta U_1$

(C) $\Delta U_2 > \Delta U_1$

(D) can not be determined

Q.19

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A diatomic gas initally at $18^{\circ}\mathrm{C}$ is compressed adiabatically to one eighth of its original volume. The temperature after compression will be

(A) $18^{\circ}\mathrm{C}$

(B) $887^{\circ}\mathrm{C}$

(C) $327^{\circ}\mathrm{C}$

(D) $395.5^{\circ}\mathrm{C}$

Q.20

Calculate the work done when 1 mole of a perfect gas is compressed adiabatically. The initial pressure and volume of the gas are $10^5\ \mathrm{N/m^2}$ and 6 litre respectively. The final volume of the gas is 2 litres. Molar specific heat of the gas at constant volume is 3R/2. [Given $(3)^{5/3} = 6.19$]

(A) $-957\ \mathrm{J}$

(B) $+957\ \mathrm{J}$

(C) $-805\ \mathrm{J}$

(D) $+805\ \mathrm{J}$

Answers & Solutions

Q.1 Answer:

$44.1\ \mathrm{kJ\ mol^{-1}}$

Solution:

Given

$\mathrm{H_2(\ g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l)}$

$\Delta \mathrm{H^{\circ}} = -285.9\ \mathrm{kJ\ mol^{-1}} \quad \ldots$

$\mathrm{H_2(\ g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(g)}$

$\Delta \mathrm{H^{\circ}} = -241.8\ \mathrm{kJ\ mol^{-1}} \quad \ldots$

We have to calculate

$\mathrm{H_2O(l) \longrightarrow H_2O(g)}; \Delta \mathrm{H^{\circ}} = ?$

On substracting eqn. (2) from eqn. (1) we get $\mathrm{H_2O(l) \longrightarrow H_2O(g)}$

$\Delta \mathrm{H^{\circ}} = -241.8 - (-285.9)$

$= 44.1\ \mathrm{kJ\ mol^{-1}}$

Q.2 Answer:

$-328\ \mathrm{kJ\ mol^{-1}}$

Solution:

Applying Hess's Law

$\Delta_{\mathrm{f}}\mathrm{H^{\circ}} = \Delta_{\mathrm{sub}}\mathrm{H} + \frac{1}{2}\Delta_{\mathrm{diss}}\mathrm{H} + \textbf{I.E.} + \textbf{E.A} + \Delta_{\mathrm{lattice}}\ \mathrm{H}$

$-617 = 161 + 520 + 77 + \textbf{E.A.} + (-1047)$

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E.A. $= -617 + 289 = -328 \text{ kJ mol}^{-1}$

$\therefore$ **electron affinity of fluorine** $= -328 \text{ kJ mol}^{-1}$

Q.3 Answer:

Cl^+ has lower hydration enthalpy

Solution:

Hydration energy of Cl^+ **is very less than** H^+**, hence it doesn't form** Cl^+ **(aq) ion.**

Q.4 Answer:

-105 kJ/mol

Solution:

Given that

$MgSO_4(s) + nH_2O \rightarrow MgSO_4nH_2O;$

$\Delta_r H_1 = -91.2 \text{ kJ/mol}$

$MgSO_4.7H_2O(s) + (n-7)H_2O$

$\rightarrow MgSO_4\,(nH_2O)$

$\Delta_r H_2 = 13.8 \text{ kJ/mol}$

or $\Delta H_{hyd} = \Delta_r H_1 - \Delta_r H_2$ **equation (i) − (ii)**

$= -91.2 \text{ kJ/mol} - 13.8 \text{ kJ/mol}$

$= -105 \text{ kJ/mol}$

Q.5 Answer:

702.04 kJ

Solution:

$$CaO(s) + CO_2(g) - CaCO_3(s)$$

$$\Delta_f H^\circ = \Delta H_f^\circ(CaCO_3) - \Delta H_f^\circ(CaO) - \Delta H_f^\circ(CaCO_2)$$

$$= -1207 - (-635) - (-394)$$

$$= -178 \text{ kJ/mol}$$

$$\therefore \quad \Delta U = \Delta H - \Delta n_g RT$$

$$\Delta U = -178 - \left(\frac{(-1) \times 8.3 \times 300}{1000}\right)$$

$$= -175.51 \text{ kJ}$$

$$n_{CaO} = \frac{224}{56} = 4$$

$$\therefore \quad q_v = n\Delta r = 4 \times (-175.51)$$

$$= -702.04 \text{ kJ}$$

Q.6 Answer:

$-100x$

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Solution:

In question 0.01 gev of NaOH is being neutralised by 0.01 gev of HCl and value is xkJ, for 1 gev the value is $-100\mathrm{x}$ (exothermic process).

Q.7 Answer:

$+64.8\mathrm{kcalmol}^{-1}$

Solution:

As $\Delta_f H$ increases stability decreases.

Q.8 Answer:

x

Solution:

Enthalpy of neutralisation of HCl with NaOH is x. In question gev of HCl $\left(\frac{500\times2}{1000}=\mathrm{lgev}\right)$ and NaOH $\left(\frac{250\times4}{1000}=\mathrm{lg\,ev}\right)$
hence the value x.

Q.9 Answer:

$+4.44\ \mathrm{kJ\ mol}^{-1}$

Solution:

$$\underset{\text{Strong acid}}{\mathrm{HCl}} \quad \underset{\text{(Complete ionisation)}}{\mathrm{H}^+} \quad + \mathrm{Cl}^-$$

$$\mathrm{NH_4OH} \underset{\text{Weak base}}{\rightleftharpoons} \mathrm{NH_4^+} + \mathrm{OH^-}$$

$$\Delta \mathrm{H} = x\ \mathrm{kJ\ mol}^{-1}$$

$$\mathrm{H^+ + OH^- \longrightarrow H_2O}$$

$$\Delta \mathrm{H} = -55.90\ \mathrm{kJ\ mol}^{-1}$$

(from neutralisation of strong acid and strong base) From equation (i), (ii) and (iii)

$$\mathrm{NH_4OH + HCl \longrightarrow NH_4^+ + Cl^- + H_2O}$$

$$\Delta \mathrm{H} = -51.46\ \mathrm{kJ\ mol}^{-1}$$

$$\therefore \quad x + (-55.90) = -51.46$$

$$x = -51.46 + 55.90$$

$$= 4.44\ \mathrm{kJ\ mol}^{-1}$$

$\therefore$ Enthalpy of ionisation of $\mathrm{NH_4OH}$ is $\Rightarrow 4.44\ \mathrm{kJ\ mol}^{-1}$

Q.10 Answer:

$-462\ \mathrm{kJ}$

Solution:

$$2\mathrm{Al} + \tfrac{3}{2}\mathrm{O_2} \rightarrow \mathrm{Al_2O_3}, \Delta \mathrm{H} = -1596\ \mathrm{kJ}$$

$$\ldots\ldots$$

$$2\mathrm{Cr} + \tfrac{3}{2}\mathrm{O_2} \rightarrow \mathrm{Cr_2O_3}, \Delta \mathrm{H} = -1134\ \mathrm{kJ}$$

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By(i) − (ii)

$2Al + Cr_2O_3 \rightarrow 2Cr + Al_2O_3, \Delta H = -462\ kJ$

Q.11 Answer:

$KNO_3; \Delta H^\circ = 53.5\ kJ\ mol^{-1}$

Solution:

Dissolution of KNO_3 is endothermic, hence heat is absorbed and cooling is observed.

Q.12 Answer:

$-8\ kJ\ mol^{-1}$

Solution:

Isobutane + oxygen $\rightarrow CO_2 + H_2O$

$\Delta H = -2870\ kJ\ mol^{-1} \ldots\ldots$ **(i)**

$$\text{n-butane} + \text{oxygen} \rightarrow CO_2 + H_2O$$
$$\Delta H = -2878\ kJ\ mol^{-1} \ldots . (ii)$$

(ii) $-(i)$; **n -butane** $-$ **Isobutane,**

$$\Delta H = (-2878 + 2870)$$
$$= -8\ kJ\ mol^{-1}.$$

Q.13 Answer:

$80 kcal mol^{-1}$

Solution:

Atomisation of methane

$CH_4(g) \longrightarrow C(g) + 4H(g)$

$\Delta H = 360 kcal$

$\therefore C - H$ **bond energy** $= \frac{360}{4}$

$= 90 kcal mol^{-1}$

$C_2H_6(g) \longrightarrow 2C(g) + 6H(g)$

$\Delta H = 620 kcal$

or $H_{C-C} + 6H_{C-H} = 620$

$\therefore \quad H_{C-C} = 620 - 6H_{C-H}$

$= 620 - 6 \times 90 = 80 kcal mol^{-1}$

Q.14 Answer:

$1 : 84$

Solution:

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No. of moles of O_2 required to supplied 30 kJ heat to second reaction
$= \frac{30}{1260} \times \frac{3}{2} = \frac{1}{28}$
$n_{O_2} : n_{H_2} = \frac{1}{28} : 3$ **or** $1 : 84$

Q.15 Answer:

1660 J

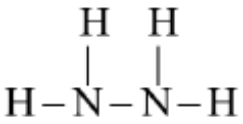
Solution:

$\mathbf{H} = E + \mathbf{PV}$
$\mathbf{H} = \mathbf{E} + \mathbf{nRT}$
$H_2 = E_2 + nRT_2$
$\mathbf{H}_1 = \mathbf{E}_1 + \mathbf{nRT}_1$
$\therefore (\mathbf{H}_2 - \mathbf{H}_1) = (\mathbf{E}_2 - \mathbf{E}_1) + \mathbf{nR}(\mathbf{T}_2 - \mathbf{T}_1)$
$\Delta H - \Delta E = \mathbf{nR}AT = 4 \times 8.3 \times 50 = 1660$ J
Thus, (d)

Q.16 Answer:

$160 KJmol^{-1}$

Solution:



(So, $4\ N - H$ **bond present)**
means their energy $= 391 \times 4 = 1564$
so the bond energy of $N - N$ **in** N_2H_4
$= 1724 - 1564 = 160 KJ/mol$

Q.17 Answer:

$T \propto \frac{1}{R}$

Solution:

As, $P = \frac{1}{3}\left(\frac{U}{V}\right)$
But $\frac{U}{V} = KT^4$
So, $P = \frac{1}{3}KT^4$
or $\frac{uRT}{V} = \frac{1}{3}KT^4 [AsPV = uRT]$
$\frac{4}{3}pR^3\ T^3 =$ **constant**
Therefore, $T \propto \frac{1}{R}$

Q.18 Answer:

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$\Delta U_1 = \Delta U_2$

Solution:

Change in internal energy do not depend upon the path followed by the process. It only depends on initial and final states i.e., $\Delta U_1 = \Delta U_2$

Q.19 Answer:

$395.5^\circ C$

Solution:

$T_1 = 18^\circ C = (273 + 18) = 291\ K$

and $V_2 = V_1/8$

We know that $TV^{\gamma-1} =$ **constant**

or, $T_2\ V_2^{\gamma-1} = T_1\ V_1^{\gamma-1}$

$\therefore \quad T_2 = T_1 \left(\frac{V_1}{V_2}\right)^{\gamma-1} = 291 \times (8)^{1.4-1}$

$= 668.5\ K = 395.5^\circ C$

Q.20 Answer:

$-957\ J$

Solution:

For an adiabatic change $PV^\gamma =$ **constant** $P_1\ V_1^\gamma = P_2\ V_2^\gamma$

As molar specific heat of gas at constant volume

$$C_v = \frac{3}{2}R$$

$$C_P = C_V + R = \frac{3}{2}R + R = \frac{5}{2}R$$

$$\gamma = \frac{C_P}{C_V} = \frac{(5/2)R}{(3/2)R} = \frac{5}{3}$$

$\therefore$ From eq$^n \cdot (1)$

$$P_2 = \left(\frac{V_1}{V_2}\right)^\gamma P_1 = \left(\frac{6}{2}\right)^{5/3} \times 10^5\ N/m^2$$

$$= (3)^{5/3} \times 10^5 = 6.19 \times 10^5\ N/m^2$$

Work done

$= \frac{1}{1-(5/3)} \left[6.19 \times 10^5 \times 2 \times 10^{-3} - 10^{-5} \times 6 \times 10^{-3}\right]$

$= -\left[\frac{2 \times 10^2 \times 3}{2}(6.19 - 3)\right]$

$= -3 \times 10^2 \times 3.19 = -957$ **joules**

[-ve sign shows external work done on the gas]

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